

IOWA STATE UNIVERSITY

LAB 7 - ADHESIVE JOINTS REPORT

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Aer E 4260 Design of Aerospace Structures

AMES, OCTOBER 2024

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INTRODUCTION

This report focuses on the design and analysis of adhesive joints using AF163_2 adhesive from 3M. The objective is to evaluate the performance of both double lap joints and scarf joints under specified loading conditions. The adhesive is available in various thicknesses (2.5, 5.5, 7.5, and 9.5 mils), and each joint configuration must support a load of 5000 lbs over a 1-inch-wide strip. For each adhesive thickness, the practical design joint length, joint strength, shear factor of safety, and peel stress are calculated.

Additionally, we analyze how joint performance changes with temperature, comparing results at room temperature (75°F) and elevated temperature (180°F). Using the same adhesive properties, a scarf joint is also designed to determine the angle of the joint and the load it can sustain.

FIGURES AND TABLES

B. Thick Adherend Shear Properties – 3M Test Method C-288

Cure Cycle: 270°F, 60 minutes, 40 psi, 1°F/min.

Adherends: 2024 T-3 Bare 0.25" thick aluminum – FPL etched

Primer: 3M™ Scotch-Weld™ Structural Adhesive Primer EC-3924B

Test Temp.	Ultimate Shear (psi)	Ultimate Elong. (in)	Yield Stress (psi)	Yield Elong. (in)	Shear Modulus (psi)
75°F	6950	.0052	5255	.00074	63,685
180°F	5780	.0114	3075	.00079	26,495

Table 2.1: Thick Adherend Shear Properties

Tensile Strength and Modulus (ASTM D 638)

Cured: Free Film Strips approx. 1/4" x 3" x .01" thick

Cure: 90 minutes at 235°F

Temperature	Ultimate Strength (psi)	Modulus (psi)
-67°F	11,000	2.3×10^5
75°F	7,000	1.6×10^5
180°F	3,000	6×10^4

Table 2.2: Tensile Strength and Modulus

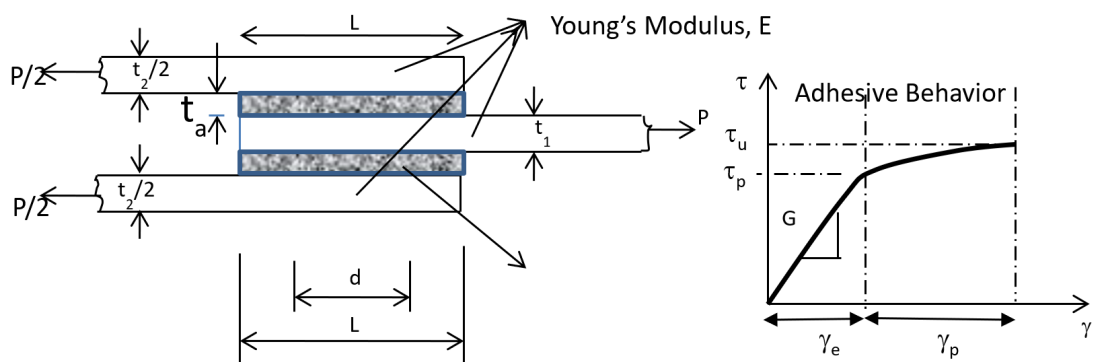


Figure 2.1: Side view of joint and shear stress vs. strain graph

METHODOLOGY AND RESULTS

3.1 Problem 1

For problem 1, we are prompted to design a double lap joint using a AF163_2 adhesive from 3M at 75°F. There are 4 thicknesses available for the adhesive $t_a = 2.5, 5.5, 7.5$ and 9.5 mils (10^{-3} in). Thus, I performed the calculations described next in a MATLAB code to generate an output for each thickness.

The first step is to calculate the practical design joint length, L_{P-D} . From the problem statement, we know the adherend Young's modulus $E_0 = 10^7$ psi, its thicknesses $t_1 = t_2 = 0.25$ in, and the applied load of 5000 lbs over a width of 1 in, thus $P = 5000$ lbs/in. From Table 2.1, we know $\tau_p = 5255$ psi and $G = 63685$ psi. We can then calculate λ and L_{P-D} .

$$\lambda = \sqrt{\frac{2(63685 \text{ psi})}{t_a} \left(\frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} + \frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} \right)} \quad (3.1)$$

$$L_{P-D} = \frac{5000 \text{ lbs/in}}{2(5255 \text{ psi})} + \frac{2}{\lambda} \quad (3.2)$$

t_a (mils)	2.5	5.5	7.5	9.5
L_{P-D} (in)	0.7890	0.9404	1.0183	1.0864

With the known values for L_{P-D} , we can proceed to calculate τ_{avg} and the joint strength. To do so, first we need to calculate γ_e and γ_p . From Table 2.1, we know $\delta_e = 0.00074$ in and $\delta_u = 0.0052$ in.

$$\gamma_e = \frac{\tau_p}{G} = \frac{5255 \text{ psi}}{63685 \text{ psi}} = 0.0825 \quad (3.3)$$

$$\gamma_u = \frac{\gamma_e \delta_u}{\delta_e} = \frac{(0.0825)(0.0052 \text{ in})}{(0.00074 \text{ in})} = 0.580 \quad (3.4)$$

$$\gamma_p = \gamma_u - \gamma_e = 0.580 - 0.0825 = 0.497 \quad (3.5)$$

$$\tau_{\text{avg}} = \frac{2}{L_{\text{P-D}} \sqrt{\frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} + \frac{1}{(10^7 \text{ psi})(0.25 \text{ in})}}} \sqrt{(5255 \text{ psi}) t_a \left(\frac{(0.0825)}{2} + (0.497) \right)} \quad (3.6)$$

$$\text{Joint Strength} = (1 \text{ in}) \tau_{\text{avg}} L_{\text{P-D}} \quad (3.7)$$

t_a (mils)	2.5	5.5	7.5	9.5
Joint Strength (lbs)	5947.94	8822.23	10302.14	11594.68

We can now also calculate the shear factor of safety with $\tau_u = 6950$ psi from [Table 2.1](#).

$$\text{FS}_{\text{shear}} = \frac{\tau_u}{\tau_{\text{avg}}} = \frac{6950 \text{ psi}}{\tau_{\text{avg}}} \quad (3.8)$$

t_a (mils)	2.5	5.5	7.5	9.5
FS_{shear}	0.92	0.74	0.69	0.65

Thus, we calculated the joint strength to be greater than the applied load of 5000 lbs for each adhesive thickness. On the other hand, our shear factors of safety are all less than 1, predicting failure for every adhesive thickness.

Moving on to peel stress calculation, we can retrieve $E_c = 1.6 \times 10^5$ psi from [Table 2.2](#). Dr. Dayal informed us we should use $\nu = 0.33$.

$$\sigma_{c_{\text{max}}} = (5255 \text{ psi}) \left(\frac{3(1.6 \times 10^5 \text{ psi})(1 - 0.33^2)(0.25 \text{ in})}{(10^7 \text{ psi}) t_a} \right)^{1/4} \quad (3.9)$$

t_a (mils)	2.5	5.5	7.5	9.5
$\sigma_{c_{\text{max}}}$ (psi)	7557.26	6205.25	5742.28	5412.76

Comparing to the ultimate strength of 7000 psi from [Table 2.2](#), we can conclude that peel won't occur for any value of t_a .

With all required values calculated, the plots were generated.

Lastly, we were prompted to calculate the practical design joint length at a temperature of 180°F. From [Table 2.1](#), we can get $G = 26495$ psi and $\tau_p = 3075$ psi for the new temperature.

$$\lambda = \sqrt{\frac{2(26495 \text{ psi})}{(2.5 \times 10^{-3} \text{ in}) \left(\frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} + \frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} \right)}} = 4.118 \text{ in}^{-1} \quad (3.10)$$

$$L_{\text{P-D}} = \frac{5000 \text{ lbs/in}}{2(3075 \text{ psi})} + \frac{2}{4.118 \text{ in}^{-1}} = 1.299 \text{ in} \quad (3.11)$$

Thus, we can conclude that $L_{\text{P-D}}$ increases as temperature increases.

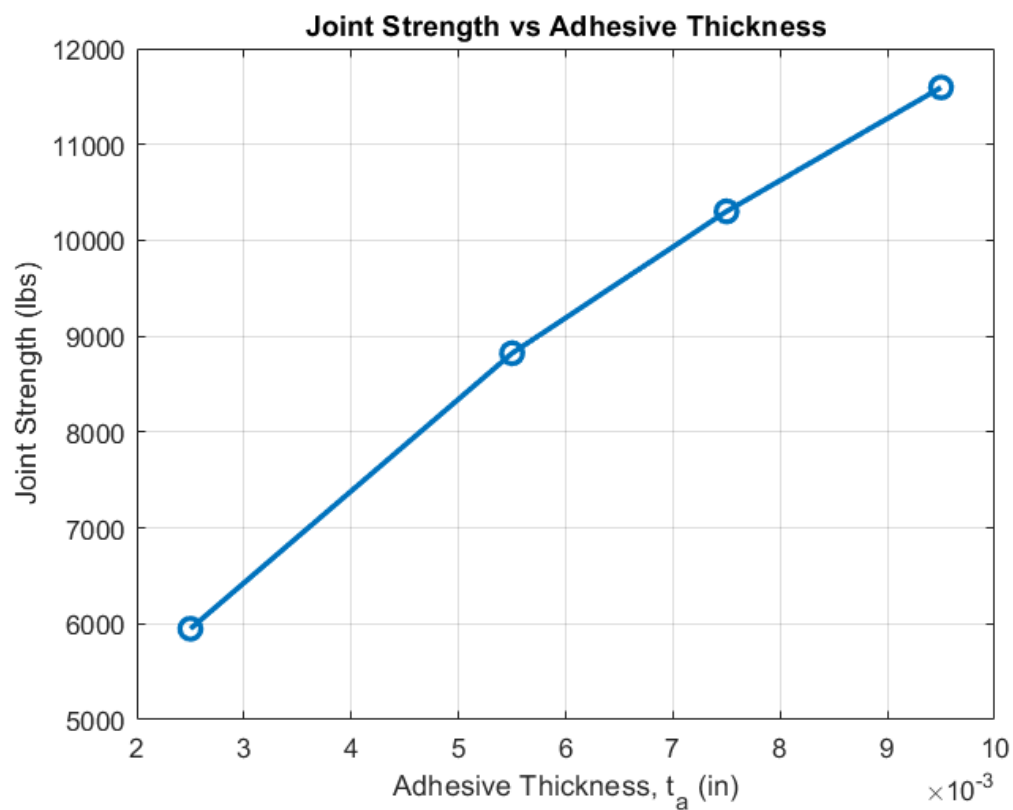


Figure 3.1: *Joint Strength vs Adhesive Thickness*

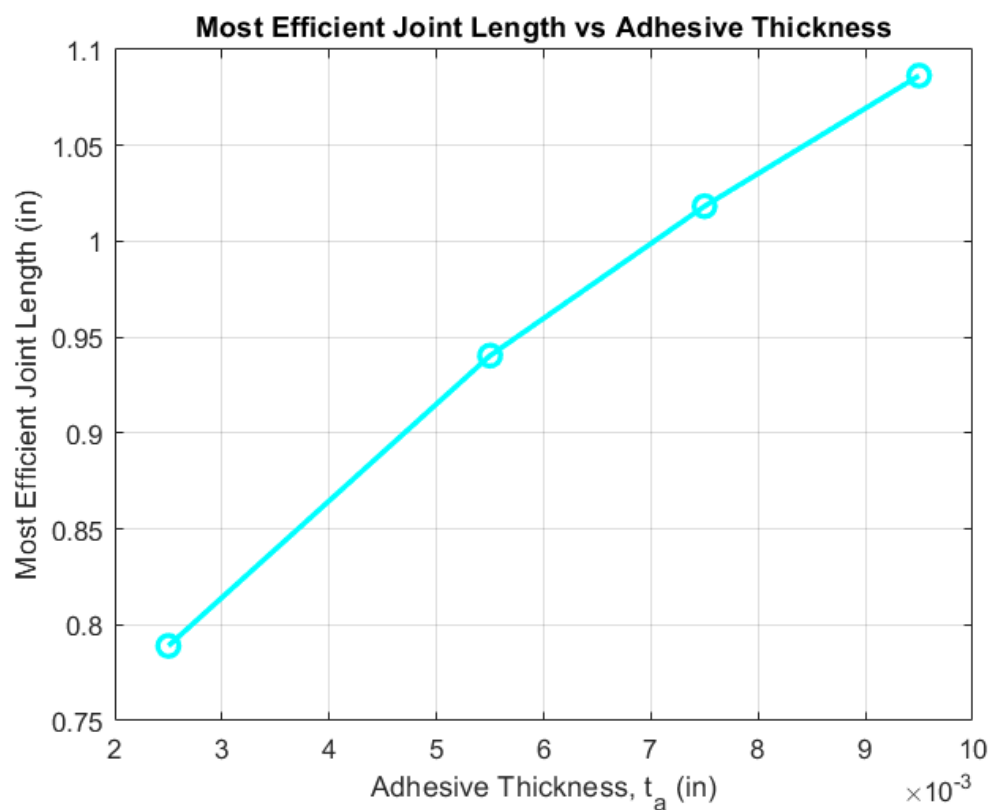


Figure 3.2: *Most Efficient Joint Length vs Adhesive Thickness*

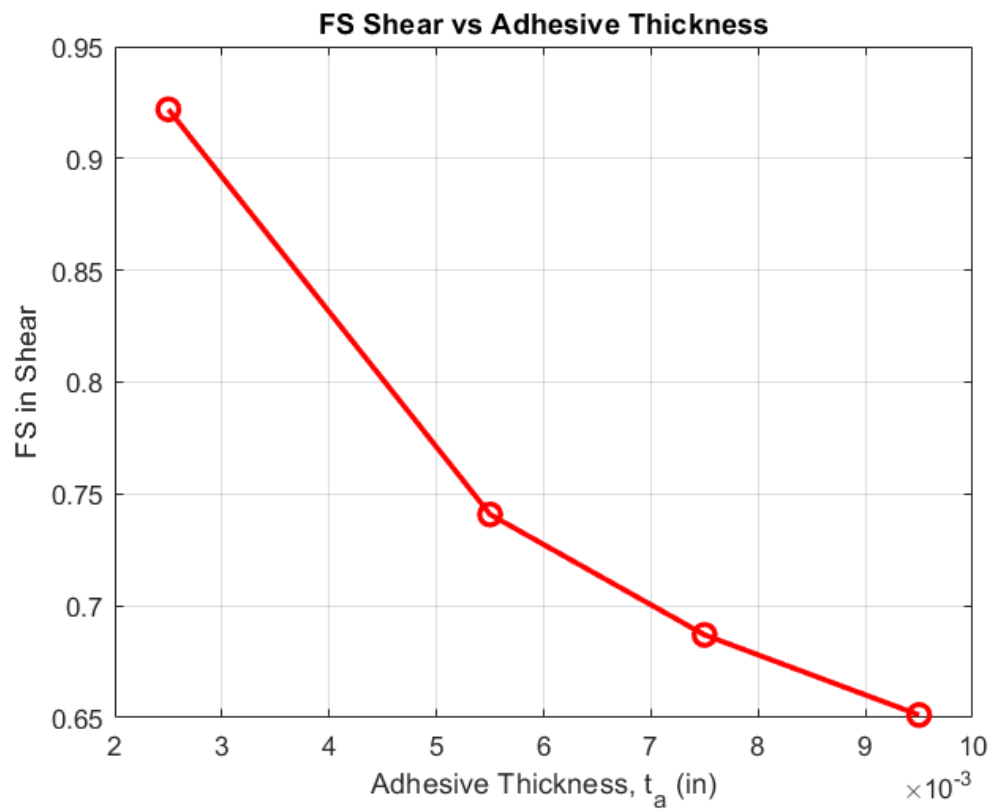


Figure 3.3: FS Shear vs Adhesive Thickness

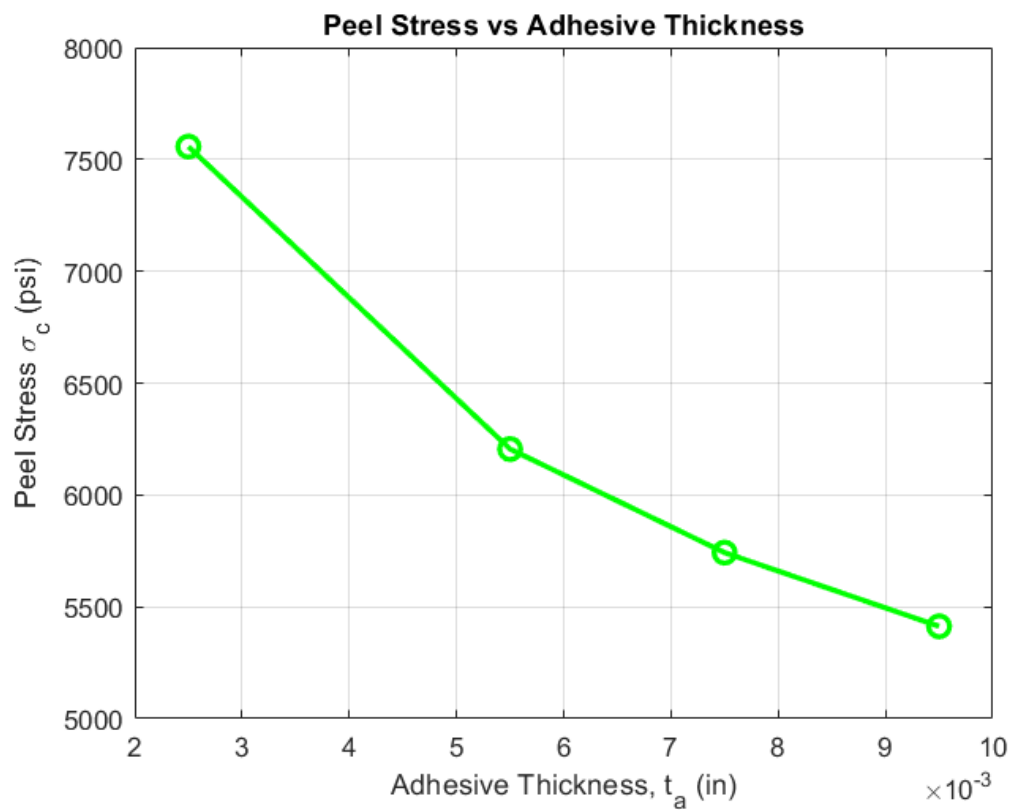


Figure 3.4: Peel Stress vs Adhesive Thickness

3.2 Problem 2

For problem 2, we were prompted to make the most efficient scarf joint for a 0.25 in aluminum plate for the same adhesive properties. θ was calculated first:

$$\theta = \tan^{-1} \frac{\sigma_{\text{ult}}}{\tau_p} = \tan^{-1} \frac{7000 \text{ psi}}{5255 \text{ psi}} = 53.104^\circ \quad (3.12)$$

Then, the load and stress which this will occur is:

$$P = \frac{\sigma_{t,u} t}{\sin^2 \theta} = \frac{(7000 \text{ psi})(0.25 \text{ in})}{\sin^2 53.104^\circ} = 2736.3 \text{ lbs/in} \quad (3.13)$$

$$\sigma_t = \frac{P}{t} = \frac{2736.3 \text{ lbs/in}}{0.25 \text{ in}} = 10.945 \text{ ksi} \quad (3.14)$$

Comparing σ_t to the ultimate strength of the aluminum plate of 40 ksi, we can conclude that the joint will fail first.

CONCLUSION

The analysis demonstrates that joint strength increases with adhesive thickness, but the shear factor of safety decreases, indicating that thicker adhesives are more prone to shear failure. Although the calculated joint strength exceeds the applied load for all thicknesses, none of the configurations achieve a shear factor of safety above 1, suggesting potential failure under shear. On the other hand, the peel stress remains below the ultimate strength of the adhesive, ensuring that peel failure is unlikely for any thickness.

At elevated temperatures (180°F), the practical design joint length increases, highlighting the impact of temperature on adhesive performance. The scarf joint design reveals that the joint will fail before the aluminum plate, emphasizing the importance of optimizing joint configuration and adhesive properties to prevent structural failure.

| A

APPENDIX A

A.1 Rough Work

AERE 426 - Lab Assignment 7

$$\textcircled{1} (a). \lambda = \sqrt{\frac{2G}{t_a} \left(\frac{1}{(Et)_1} + \frac{1}{(Et)_2} \right)} \quad t_a = 2.5 \text{ mils} = 2.5 \times 10^{-3} \text{ in}$$

$$\lambda = \sqrt{\frac{2(63685 \text{ psi})}{(2.5 \times 10^{-3} \text{ in})} \left(\frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} + \frac{1}{(10^7 \text{ psi})(0.25 \text{ in})} \right)}$$

$$\lambda = 6.384 \text{ in}^{-1}$$

$$P = 5000 \text{ lbs} \quad L_{p-d} = \frac{P}{2\tau_p \lambda} + 2 = \frac{(5000 \text{ lbs/in})}{2(5255 \text{ psi})(6.38 \text{ in}^{-1})} + 2 = 0.789 \text{ in}$$

$$(b). G = \tau_p \Rightarrow \gamma_e = \frac{\tau_p}{G} = \frac{(5255 \text{ psi})}{(63685 \text{ psi})} = 0.0825$$

$$\frac{\gamma_e}{\delta_e} = \frac{\gamma_u}{\delta_u} \Rightarrow \gamma_u = \frac{\gamma_e \delta_u}{\delta_e} = \frac{(0.0825)(.0052 \text{ in})}{(.00074 \text{ in})}$$

$$\gamma_u = 0.580$$

$$\gamma_p = \gamma_u - \gamma_e = 0.580 - 0.0825 = 0.497$$

$$\tau_{avg} = \frac{2}{(0.789 \text{ in}) \sqrt{\frac{2}{(10^7 \text{ psi})(0.25 \text{ in})}}} \sqrt{(5255 \text{ psi})(2.5 \times 10^{-3} \text{ in}) \left(\frac{\gamma_e + \gamma_p}{2} \right)}$$

$$\tau_{avg} = 7539 \text{ psi}$$

$$\text{Joint strength} = w \cdot \tau_{avg} \cdot L_{p-d} = (1 \text{ in})(7539 \text{ psi})(0.789 \text{ in})$$

$$= 5948 \text{ lbs}$$

$$(c). \text{Joint strength} > P \quad \checkmark \quad FS_{shear} = \frac{\tau_u}{\tau_{avg}} = \frac{6950 \text{ psi}}{7539 \text{ psi}} = 0.923$$

$$\begin{aligned}
 (d) \quad E_c &= 1.6 \times 10^5 \text{ psi} & \sigma_{c, \text{max}} &= \tau_p \left(\frac{3 E_c (1 - \nu_o^2) t_2}{E_o t_o} \right)^{1/4} \\
 E_o &= 10^7 \text{ psi} & &= (5255 \text{ psi})^4 \sqrt{\frac{3(1.6 \times 10^5 \text{ psi})(1 - 0.33^2)(0.25 \text{ in})}{(10^7 \text{ psi})(2.5 \times 10^{-3} \text{ in})}} \\
 \nu_o &= & &= 755.007 \text{ psi} //
 \end{aligned}$$

• Ultimate strength of 7000 psi, peel won't occur.

(e) MATLAB

$$(f) \quad \lambda_{180^\circ F} = \frac{2(26495 \text{ psi})}{\sqrt{(2.5 \times 10^{-3} \text{ in})}} \left(\frac{2}{(10^7 \text{ psi})(0.25 \text{ in})} \right)^{1/4} = 4.118 \text{ in}^{-1} //$$

$$\text{LP-D}_{180^\circ F} = \frac{(5000 \text{ lbs/in})}{2(3075 \text{ psi})} + \frac{2}{(4.118 \text{ in}^{-1})} = 1.299 \text{ in} //$$

• Increasing T increases LP-D.

$$(2) \quad \Theta = \tan^{-1} \left(\frac{\sigma_{\text{ult}}}{\tau_p} \right) = \tan^{-1} \left(\frac{7000 \text{ psi}}{5255 \text{ psi}} \right) = 0.927 \text{ rad} = 53.104^\circ //$$

$$P = \frac{\sigma_{t0} t}{\sin^2 \Theta} = \frac{(17 \times 10^3 \text{ psi})(0.25 \text{ in})}{\sin^2(53.104^\circ)} = 2736.3 \text{ lbs/in}$$

$$\sigma_t = \frac{P}{t} = \frac{2736.3 \text{ lbs/in}}{0.25 \text{ in}} = 10.945 \text{ ksi} //$$

• Since $\sigma_t < \sigma_{u, \text{al}}$, the joint will fail first.

APPENDIX B

B.1 MATLAB Code

```
1 % AerE 426, Lab Assignment 7 - Adhesive Joint: Problem 1, Lucas Tavares
2
3 clear, clc, close all
4
5 % Adhesive thicknesses
6 ta = [2.5e-3, 5.5e-3, 7.5e-3, 9.5e-3]; %in
7
8 % Strips thicknesses
9 t1 = 0.25; %in
10 t2 = 0.25; %in
11
12 % Strips Young's modulus and Poisson's ratio
13 E0 = 10e6; %psi
14 nu = 0.33;
15
16 % Strips width
17 w = 1; %in
18
19 % Adhesive properties at 75 F
20 G = 63685; %psi
21 tau_p = 5255; %psi
22 delta_u = 0.0052; %in
23 delta_e = 0.00074; %in
24 tau_u = 6950; %psi
25 Ec = 1.6e5; %psi
26
27 %Strain calculations
28 gamma_e = tau_p / G;
29 gamma_u = gamma_e * delta_u / delta_e;
30 gamma_p = gamma_u - gamma_e;
31
32 % Load applied
33 P = 5000; %lbs/in
34
35 % Calculations for tau_avg and joint strength
```

```

36 lambda = sqrt((2 * G ./ ta) .* ((1 / (E0 * t1)) + (1 / (E0 * t2)))); %in^-1
37 L_practical_design = (P / (2 * tau_p)) + (2 ./ lambda); %in
38 tau_avg = (2 ./ (L_practical_design .* sqrt((1 / (E0 * t1)) + (1 / (E0 * t2)
    )))) .* sqrt(tau_p .* ta .* ((gamma_e / 2) + gamma_p)); %psi
39 joint_strength = w .* tau_avg .* L_practical_design; %lbs
40
41 % Calculation for FS in shear
42 FS_shear = tau_u ./ tau_avg;
43
44 % Calculation for peel stress
45 sigma_c_max = tau_p .* (3 * Ec * (1 - nu^2) * t2 ./ (E0 .* ta)).^(1 / 4);
46
47 % Print results
48 fprintf('\nResults for each adhesive thickness (ta):\n');
49 fprintf('-----\n');
50 fprintf('ta (in) | L_design (in) | Strength (lbs) | FS_shear | Peel Stress (
    psi)\n');
51 fprintf('-----\n');
52 for i = 1:length(ta)
53     fprintf('%4f | %4f | %2f | %2f | %2f\n', ...
54         ta(i), L_practical_design(i), joint_strength(i), FS_shear(i),
55         sigma_c_max(i));
56 end
57 fprintf('-----\n\n');
58
59 % Create plots
60 figure(1)
61 plot(ta, joint_strength, '-o', 'LineWidth', 2, 'MarkerSize', 8)
62 xlabel('Adhesive Thickness, t_a (in)')
63 ylabel('Joint Strength (lbs)')
64 title('Joint Strength vs Adhesive Thickness')
65 grid on
66
67 % Create plots
68 figure(2)
69 plot(ta, L_practical_design, '-o', 'LineWidth', 2, 'MarkerSize', 8, 'Color',
    'c')
70 xlabel('Adhesive Thickness, t_a (in)')
71 ylabel('Most Efficient Joint Length (in)')
72 title('Most Efficient Joint Length vs Adhesive Thickness')
73 grid on
74
75 figure(3)
76 plot(ta, FS_shear, '-o', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'r')
77 xlabel('Adhesive Thickness, t_a (in)')
78 ylabel('FS in Shear')
79 title('FS Shear vs Adhesive Thickness')
80 grid on
81
82 figure(4)
83 plot(ta, sigma_c_max, '-o', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'g')

```



```
83 xlabel('Adhesive Thickness, t_a (in)')
84 ylabel('Peel Stress \sigma_c (psi)')
85 title('Peel Stress vs Adhesive Thickness')
86 grid on
```